

Q1

Angle bisector can be $\vec{a} = \lambda (\hat{b} + \hat{c})$ or $\vec{a} = \mu (\hat{b} - \hat{c})$

$$\vec{a} = \lambda \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} + \frac{\hat{i} + \hat{j} + 4\hat{k}}{3\sqrt{2}} \right) = \frac{\lambda}{3\sqrt{2}} [3\hat{i} + 3\hat{j} + \hat{i} - \hat{j} + 4\hat{k}]$$

$$= \frac{\lambda}{3\sqrt{2}} [4\hat{i} + 2\hat{j} + 4\hat{k}]$$

Compare with $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}$

$$\frac{2\lambda}{3\sqrt{2}} = 2 \Rightarrow \lambda = 3\sqrt{2}$$

$$\vec{a} = 4\hat{i} + 2\hat{j} + 4\hat{k}$$

Not in option so now consider $\vec{a} = \mu \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} - \frac{\hat{i} - \hat{j} + 4\hat{k}}{3\sqrt{2}} \right)$

$$\vec{a} = \frac{\mu}{3\sqrt{2}} (3\hat{i} + 3\hat{j} - \hat{i} + \hat{j} - 4\hat{k})$$

$$= \frac{\mu}{3\sqrt{2}} (2\hat{i} + 4\hat{j} - 4\hat{k})$$

Compare with $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}$

$$\frac{4\mu}{3\sqrt{2}} = 2 \Rightarrow \mu = \frac{3\sqrt{2}}{2}$$

$$\vec{a} = \hat{i} + 2\hat{j} - 2\hat{k}$$

$$\vec{a} \cdot \hat{k} + 2 = 0$$

$$-2 + 2 = 0$$

Q2

$$|\vec{a} + \vec{b} + \vec{c}|^2 = 0$$

$$3 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0$$

$$(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = \frac{-3}{2}$$

$$\Rightarrow \lambda = \frac{-3}{2}$$

$$\vec{d} = \vec{a} \times \vec{b} + \vec{b} \times (-\vec{a} - \vec{b}) + (-\vec{a} - \vec{b}) \times \vec{a}$$

$$= \vec{a} \times \vec{b} + \vec{a} \times \vec{b} + \vec{a} \times \vec{b}$$

$$\vec{d} = 3(\vec{a} \times \vec{b})$$

Q3

$$\pm 1 = \begin{vmatrix} 1 & 1 & \lambda \\ 1 & 1 & 3 \\ 2 & 1 & 1 \end{vmatrix} \Rightarrow -\lambda + 3 = \pm 1$$

$$\Rightarrow \lambda = 2 \text{ or } \lambda = 4$$

For $\lambda = 4$

$$\cos \theta = \frac{2+1+4}{\sqrt{6}\sqrt{18}} = \frac{7}{6\sqrt{3}}$$

Q4

$$\vec{a} \times (\vec{b} \times \vec{c}) = \vec{a} \times (\vec{b} \times \vec{a})$$

$$- (\vec{a} \cdot \vec{b}) \vec{c} = (\vec{a} \cdot \vec{a}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{a}$$

$$-4\vec{c} = 6(\hat{i} - \hat{j} + \hat{k}) - 4(\hat{i} - 2\hat{j} + \hat{k})$$

$$-4\vec{c} = 2\hat{i} + 2\hat{j} + 2\hat{k}$$

$$\vec{c} = -\frac{1}{2}(\hat{i} + \hat{j} + \hat{k})$$

$$\vec{b} \cdot \vec{c} = -\frac{1}{2}$$

Q5

$$\begin{vmatrix} a+1 & a & a \\ a & a+1 & a \\ a & a & a+1 \end{vmatrix} = 0 \Rightarrow a+1+a+a=0$$

$$\Rightarrow a = -\frac{1}{3}$$

$$\vec{P} = \frac{2}{3}\hat{i} - \frac{1}{3}\hat{j} - \frac{1}{3}\hat{k}$$

$$\vec{Q} = \frac{1}{3}(-\hat{i} + 2\hat{j} - \hat{k})$$

$$\vec{R} = \frac{1}{3}(-\hat{i} - \hat{j} + 2\hat{k})$$

$$\vec{P} \cdot \vec{Q} = \frac{1}{9}(-2 - 2 + 1) = -\frac{1}{3}$$

$$\vec{R} \times \vec{Q} = \frac{1}{9} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{vmatrix}$$

$$= \frac{1}{9}(\hat{i}(4-1) - \hat{j}(-2-1) + \hat{k}(1+2))$$

$$= \frac{1}{3}(3\hat{i} + 3\hat{j} + 3\hat{k}) = \frac{\hat{i} + \hat{j} + \hat{k}}{3}$$

$$|\vec{R} \times \vec{Q}| = \frac{1}{3}\sqrt{3} \Rightarrow |\vec{R} \times \vec{Q}|^2 = \frac{1}{3}$$

$$3(\vec{P} \cdot \vec{Q})^2 - \lambda |\vec{R} \times \vec{Q}|^2 = 0$$

$$3 \cdot \frac{1}{9} - \lambda \cdot \frac{1}{3} = 0 \Rightarrow \lambda = 1$$

Q6

$$\vec{AB} = (\hat{i}) - (3\hat{j}) + 8\hat{k}$$

$$\vec{CD} = 4\hat{i} - 4\hat{j} + 7\hat{k}$$

$$\left(\frac{\vec{AB} \cdot \vec{CD}}{\sqrt{16 + 16 + 49}} \right) = \frac{4 + 12 + 56}{\sqrt{81}} = \frac{72}{9} = 8$$

Q7

Given, $|\vec{a}| = |\vec{b}| = |\vec{c}| = 1$

Also $|\vec{a} - \vec{b}|^2 + |\vec{a} - \vec{c}|^2 = 8$

$$\Rightarrow |\vec{a}|^2 + |\vec{b}|^2 - 2\vec{a} \cdot \vec{b} + |\vec{a}|^2 + |\vec{c}|^2 - 2\vec{a} \cdot \vec{c} = 8$$

$$\Rightarrow \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} = -2$$

Now, $|\vec{a} + 2\vec{b}|^2 + |\vec{a} + 2\vec{c}|^2$

$$= |\vec{a}|^2 + 4|\vec{b}|^2 + 4\vec{a} \cdot \vec{b} + |\vec{a}|^2 + 4|\vec{c}|^2 + 4\vec{a} \cdot \vec{c}$$

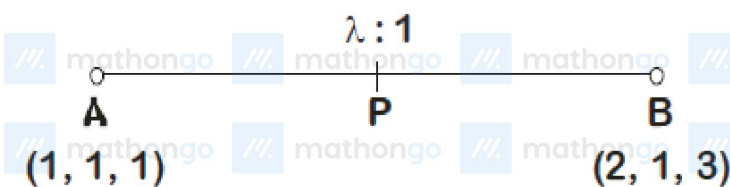
$$= 10 + 4(\vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c})$$

$$= 10 + 4(-2)$$

$$= 2$$

Q8

Let position vector of P is $\overrightarrow{OP} = \frac{\lambda\vec{b} + \vec{a}}{\lambda + 1}$



Given $\overrightarrow{OB} \cdot \overrightarrow{OP} - 3|\overrightarrow{OA} \times \overrightarrow{OP}|^2 = 6$

$$\Rightarrow \vec{b} \cdot \left(\frac{\lambda\vec{b} + \vec{a}}{\lambda + 1} \right) - 3 \left| \vec{a} \times \frac{\lambda\vec{b} + \vec{a}}{\lambda + 1} \right|^2 = 6$$

$$\Rightarrow \frac{\vec{a} \cdot \vec{b} + \lambda|\vec{b}|^2}{\lambda + 1} - \frac{3\lambda^2}{(\lambda + 1)^2} |\vec{a} \times \vec{b}|^2 = 6$$

$$\Rightarrow \frac{6 + 14\lambda}{\lambda + 1} - \frac{18\lambda^2}{(\lambda + 1)^2} = 6$$

$$\Rightarrow 6 + \frac{8\lambda}{\lambda + 1} - \frac{18\lambda^2}{(\lambda + 1)^2} = 6$$

Let $\frac{\lambda}{\lambda + 1} = t$

$$18t^2 - 8t = 0$$

$$t = 0, \frac{4}{9}$$

$$\therefore \frac{\lambda}{\lambda + 1} = \frac{4}{9}$$

$$\therefore \lambda = \frac{4}{5} = 0.8$$

Q9

$$\begin{aligned} \therefore a \cos \theta &= b \cos \left(\theta + \frac{2\pi}{3} \right) = c \cos \left(\theta + \frac{4\pi}{3} \right) = k \\ ab + bc + ca &= k^2 \left[\frac{1}{\cos \theta \cdot \cos \left(\theta + \frac{2\pi}{3} \right)} + \frac{1}{\cos \left(\theta + \frac{2\pi}{3} \right) \cdot \cos \left(\theta + \frac{4\pi}{3} \right)} + \frac{1}{\cos \left(\theta + \frac{4\pi}{3} \right) \cdot \cos \theta} \right] \\ &= k^2 \left[\frac{\cos \theta + \cos \left(\theta + \frac{2\pi}{3} \right) + \cos \left(\theta + \frac{4\pi}{3} \right)}{\cos \theta \cdot \cos \left(\theta + \frac{2\pi}{3} \right) \cdot \cos \left(\theta + \frac{4\pi}{3} \right)} \right] \\ &= 0 \end{aligned}$$

Q10

$$\begin{aligned} \text{Here } \vec{a} \cdot (\vec{b} \times \vec{c}) &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & -2 & 3 \\ -2 & x & -1 \\ 7 & -2 & x \end{vmatrix} = x^3 - 27x + 26 \\ \Rightarrow f(x) &= x^3 - 27x + 26 \\ f'(x) &= 3x^2 - 27 = 0 \Rightarrow x = -3, 3 \\ &+ \frac{-1}{-3} \frac{1}{3} \\ \text{Max Min} \end{aligned}$$

$$\Rightarrow x_0 = -3$$

$$\text{Now } \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -2x - 2x - 3 - 14 - 2x - x + 7x + 4 + 3x = 3x - 13$$

$$\text{So value at } x = x_0 = 3x - 3 - 13 = -22$$

Q11

$$\text{Let } \vec{a} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\text{Now } \hat{i} \times (\vec{a} \times \hat{i}) = (\hat{i} \cdot \hat{i})\vec{a} - (\hat{i} \cdot \vec{a})\hat{i}$$

$$= y\hat{j} + z\hat{k}$$

$$\text{similarly } \hat{j} \times (\vec{a} \times \hat{j}) = x\hat{i} + z\hat{k}$$

$$\hat{k} \times (\vec{a} \times \hat{k}) = x\hat{i} + y\hat{j}$$

$$\text{Now } |y\hat{j} + z\hat{k}|^2 + |x\hat{i} + z\hat{k}|^2 + |x\hat{i} + y\hat{j}|^2$$

$$= 2(x^2 + y^2 + z^2) = 2(4 + 1 + 4) = 18$$

Q12

Volume of parallelopiped $= [\vec{a} \ \vec{b} \ \vec{c}]$

$$\begin{vmatrix} 1 & 1 & n \\ 2 & 4 & -n \\ 1 & n & 3 \end{vmatrix} = 158$$

$$\Rightarrow (12 + n^2) - 1(6 + n) + n(2n - 4) = 158$$

$$\Rightarrow 3n^2 - 5n - 152 = 0$$

$$\Rightarrow 3n^2 - 24n + 19n - 152 = 0$$

$$\Rightarrow 3n(n - 8) + 19(n - 8) = 0$$

$$\Rightarrow n = 8$$

$$\therefore \vec{a} = \hat{i} + \hat{j} + 8\hat{k}, \vec{b} = 2\hat{i} + 4\hat{j} - 8\hat{k} \text{ and } \vec{c} = \hat{i} + 8\hat{j} + 3\hat{k}$$

$$\vec{a} \cdot \vec{c} = 1 + 8 + 24 = 33$$

$$\vec{b} \cdot \vec{c} = 2 + 32 - 24 = 10$$

Q13

Projection of $\vec{b}$ on $\vec{a}$ = Projection of $\vec{c}$ on $\vec{a}$

$$\Rightarrow \vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c}$$

$$\text{Given } \vec{b} \cdot \vec{c} = 0$$

$$\therefore |\vec{a} + \vec{b} - \vec{c}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2\vec{a} \cdot \vec{b} - 2\vec{b} \cdot \vec{c} - 2\vec{a} \cdot \vec{c}$$

$$= 4 + 16 + 16$$

$$= 36$$

$$\Rightarrow |\vec{a} + \vec{b} - \vec{c}| = 6$$

Q14

Let angle between $\vec{a}$ and $\vec{b}$ be θ

$$|\vec{a} + \vec{b}| = \sqrt{1 + 1 + 2 \cos \theta} = 2 \left| \cos \frac{\theta}{2} \right|$$

$$\text{and } |\vec{a} - \vec{b}| = 2 \left| \sin \frac{\theta}{2} \right|$$

$$\text{So, } \sqrt{3}|\vec{a} + \vec{b}| + |\vec{a} - \vec{b}| = 2 \left[\sqrt{3} \left| \cos \frac{\theta}{2} \right| + \left| \sin \frac{\theta}{2} \right| \right]$$

$$\max \{ \sqrt{3}|\vec{a} + \vec{b}| + |\vec{a} - \vec{b}| \} = 2 \sqrt{(\sqrt{3})^2 + (1)^2}$$

$$= 4$$

Q15

$$|\vec{x} + \vec{y}| = |\vec{x}|$$

Squaring both sides we get

$$|\vec{x}|^2 + 2\vec{x} \cdot \vec{y} + |\vec{y}|^2 = |\vec{x}|^2$$

$$\Rightarrow 2\vec{x} \cdot \vec{y} + \vec{y} \cdot \vec{y} = 0$$

Also $2\vec{x} + \lambda\vec{y}$ and $\vec{y}$ are perpendicular

$$\therefore 2\vec{x} \cdot \vec{y} + \lambda\vec{y} \cdot \vec{y} = 0$$

$$\text{Comparing (1) \& (2) } \lambda = 1$$

Q1 JEE Main 2020 - 7 Jan (Morning)

A vector $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}$ ($\alpha, \beta \in R$) lies in the plane of the vectors, $\vec{b} = \hat{i} + \hat{j}$ and $\vec{c} = \hat{i} - \hat{j} + 4\hat{k}$. If $\vec{a}$ bisects the angle between $\vec{b}$ and $\vec{c}$, then:

- (A) $\vec{a} \cdot \hat{k} + 2 = 0$
- (B) $\vec{a} \cdot \hat{k} + 4 = 0$
- (C) $\vec{a} \cdot \hat{k} - 2 = 0$
- (D) $\vec{a} \cdot \hat{k} + 5 = 0$

Q2 JEE Main 2020 - 7 Jan (Evening)

Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$. If $\lambda = \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$ and $\vec{d} = \vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}$, then the ordered pair, $(\lambda, \vec{d})$ is equal to :

- (A) $\left(\frac{3}{2}, 3\vec{a} \times \vec{b}\right)$
- (B) $\left(\frac{3}{2}, 3\vec{a} \times \vec{c}\right)$
- (C) $\left(-\frac{3}{2}, 3\vec{a} \times \vec{c}\right)$
- (D) $\left(-\frac{3}{2}, 3\vec{a} \times \vec{b}\right)$

Q3 JEE Main 2020 - 8 Jan (Morning)

Let the volume of a parallelepiped whose coterminous edges are given by $\vec{u} = \hat{i} + \hat{j} + \lambda\hat{k}$, $\vec{v} = \hat{i} + \hat{j} + 3\hat{k}$ and $\vec{w} = 2\hat{i} + \hat{j} + \hat{k}$ be 1 cu. unit. If θ be the angle between the edges $\vec{u}$ and $\vec{w}$, then $\cos \theta$ can be :

- (A) $\frac{7}{3\sqrt{10}}$
- (B) $\frac{7}{6\sqrt{3}}$
- (C) $\frac{5}{3\sqrt{3}}$
- (D) $\frac{5}{7}$

Q4 JEE Main 2020 - 8 Jan (Evening)

Let $\vec{a} = \hat{i} - 2\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ be two vectors. If $\vec{c}$ is a vector such that $\vec{b} \times \vec{c} = \vec{b} \times \vec{a}$ and $\vec{c} \cdot \vec{a} = 0$, then $\vec{c} \cdot \vec{b}$ is equal to

- (A) $\frac{1}{2}$

(B) $\frac{1}{3}$

(C) $-\frac{1}{2}$

(D) $-\frac{1}{3}$

Q5 JEE Main 2020 - 9 Jan (Morning)

If

$$\vec{P} = (a+1)\hat{i} + a\hat{j} + a\hat{k}$$

$$\vec{Q} = a\hat{i} + (a+1)\hat{j} + a\hat{k}$$

$$\vec{R} = a\hat{i} + a\hat{j} + (a+1)\hat{k}$$

and $\vec{P}$, $\vec{Q}$, $\vec{R}$ are coplanar vectors and $3(\vec{P} \cdot \vec{Q})^2 - \lambda|\vec{R} \times \vec{Q}|^2 = 0$ then value of λ is

Q6 JEE Main 2020 - 9 Jan (Morning)

The projection of the line segment joining the points $(1, -1, 3)$ and $(2, -4, 11)$ on the line joining the points $(-1, 2, 3)$ and $(3, -2, 10)$ is :

Q7 JEE Main 2020 - 2 Sep (Morning)

Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three unit vectors such that $|\vec{a} - \vec{b}|^2 + |\vec{a} - \vec{c}|^2 = 8$. Then

$|\vec{a} + 2\vec{b}|^2 + |\vec{a} + 2\vec{c}|^2$ is equal to

Q8 JEE Main 2020 - 2 Sep (Evening)

Let the position vectors of points 'A' and 'B' be $\hat{i} + \hat{j} + \hat{k}$ and $2\hat{i} + \hat{j} + 3\hat{k}$, respectively. A point 'P'

divides the line segment AB internally in the ratio $\lambda : 1 (\lambda > 0)$. If O is the origin and

$\vec{OB} \cdot \vec{OP} - 3|\vec{OA} \times \vec{OP}|^2 = 6$, then λ is equal to

Q9 JEE Main 2020 - 3 Sep (Evening)

Let $a, b, c \in \mathbb{R}$ be such that $a^2 + b^2 + c^2 = 1$. If $a \cos \theta = b \cos \left(\theta + \frac{2\pi}{3} \right) = c \cos \left(\theta + \frac{4\pi}{3} \right)$, where $\theta = \frac{\pi}{9}$, then the angle between the vectors $a\hat{i} + b\hat{j} + c\hat{k}$ and $b\hat{i} + c\hat{j} + a\hat{k}$ is :

(A) 0

(B) $\frac{\pi}{9}$

(C) $\frac{\pi}{2}$

(D) $\frac{2\pi}{3}$

Q10 JEE Main 2020 - 4 Sep (Morning)

Let x_0 be the point of local maxima of $f(x) = \vec{a} \cdot (\vec{b} \times \vec{c})$, where $\vec{a} = x\hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{b} = -2\hat{i} + x\hat{j} - \hat{k}$ and $\vec{c} = 7\hat{i} - 2\hat{j} + x\hat{k}$. Then the value of $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$ at $x = x_0$ is

- (A) 14
- (B) -30
- (C) -4
- (D) -22

Q11 JEE Main 2020 - 4 Sep (Evening)

If $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$, then the value of $|\hat{i} \times (\vec{a} \times \hat{i})|^2 + |\hat{j} \times (\vec{a} \times \hat{j})|^2 + |\hat{k} \times (\vec{a} \times \hat{k})|^2$ is equal to

Q12 JEE Main 2020 - 5 Sep (Morning)

If the volume of a parallelepiped, whose coterminal edges are given by the vectors

$\vec{a} = \hat{i} + \hat{j} + n\hat{k}$, $\vec{b} = 2\hat{i} + 4\hat{j} - n\hat{k}$ and $\vec{c} = \hat{i} + n\hat{j} + 3\hat{k}$ ($n \geq 0$), is 158 cu.units, then :

- (A) $n = 7$
- (B) $\vec{b} \cdot \vec{c} = 10$
- (C) $n = 9$
- (D) $\vec{a} \cdot \vec{c} = 17$

Q13 JEE Main 2020 - 5 Sep (Evening)

Let the vectors $\vec{a}, \vec{b}, \vec{c}$ be such that $|\vec{a}| = 2$, $|\vec{b}| = 4$ and $|\vec{c}| = 4$. If the projection of $\vec{b}$ on $\vec{a}$ is equal to the projection of $\vec{c}$ on $\vec{a}$ and $\vec{b}$ is perpendicular to $\vec{c}$, then the value of $|\vec{a} + \vec{b} - \vec{c}|$ is

Q14 JEE Main 2020 - 6 Sep (Morning)

If $\vec{a}$ and $\vec{b}$ are unit vectors, then the greatest value of $\sqrt{3}|\vec{a} + \vec{b}| + |\vec{a} - \vec{b}|$ is

Q15 JEE Main 2020 - 6 Sep (Evening)

If $\vec{x}$ and $\vec{y}$ be two non-zero vectors such that $|\vec{x} + \vec{y}| = |\vec{x}|$ and $2\vec{x} + \lambda\vec{y}$ is perpendicular to $\vec{y}$, then the value of λ is

Answer Key

Q1 (A)	Q2 (D)	Q3 (B)	Q4 (C)
Q5 (1)	Q6 (8)	Q7 (2)	Q8 (0.8)
Q9 (D)	Q10 (D)	Q11 (18)	Q12 (B)
Q13 (6)	Q14 (4)	Q15 (1)	